NAME	:	M. Adi lakshmi
VTU	:	29837
COURSE NAME	:	Problem solving and testing using java
COURSE CODE	:	10211CS227

Task 3 (Easy): Maximum Profit Analyzer

PROBLEM DESCRIPTION:

Given daily profit/loss values, find the maximum possible profit obtainable from a contiguous sequence of days using Kadane’s Algorithm.

Input Format

- First line contains integer N.
- Second line contains N integers.

Output Format

Print maximum subarray sum.

Constraints

- 1 ≤ N ≤ 10^5

Sample Input

8

-2 -3 4 -1 -2 1 5 -3

Sample Output

7

PROGRAM:

```
import java.util.Scanner;


public class Main {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);


    int N = sc.nextInt();
    int[] arr = new int[N];


    for (int i = 0; i < N; i++) {
      arr[i] = sc.nextInt();
```

```
        }

        int maxEndingHere = arr[0];
        int maxSoFar = arr[0];

        for (int i = 1; i < N; i++) {
            maxEndingHere = Math.max(arr[i], maxEndingHere + arr[i]);
            maxSoFar = Math.max(maxSoFar, maxEndingHere);
        }
        System.out.println(maxSoFar);
    }
}
```

OUTPUT:

```
8
-2 -3 4 -1 -2 1 5 -3
7
```